

SAP-SLAM: Semantic-Assisted Perception SLAM With 3D Gaussian Splatting

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Abstract—The integration of 3D Gaussians has introduced a novel scene representation in Simultaneous Localization and Mapping (SLAM), characterized by explicit representation and differentiable rendering capabilities that enhance scene reconstruction and understanding. However, most current SLAM systems only exploit the basic representational capacity of 3D Gaussians, neglecting their potential to offer richer information and facilitate higher-dimensional scene comprehension. Furthermore, these systems often struggle with reconstruction when encountering rapid camera movements or depth missing. Drawing inspiration from 3D language field, which explores the intrinsic relationships among scene objects, we propose SAP-SLAM, a dense SLAM system that combines high-fidelity reconstruction and advanced semantic understanding. Our approach leverages pre-trained visual models to extract semantic features, which are then fused, dimensionally reduced, and encoded into the 3D Gaussian model for optimization and rendering. The integration of these features improves the systems semantic comprehension and scene representation, ultimately enabling the creation of high-precision 3D semantic maps. Additionally, we introduce a semantic-guided Gaussian densification and pruning strategy, which uses semantic consistency to prioritize attention on poorly reconstructed areas, greatly improving performance in complex scenarios. SAP-SLAM achieves competitive results on both real-world and synthetic datasets, demonstrating superior capabilities in semantic understanding and reconstruction.

I. INTRODUCTION

Simultaneous Localization and Mapping (SLAM) is an essential capability for robots navigating and exploring unknown environments. SLAM allows robots to simultaneously update maps and track the location of an agent. Its advancement has greatly facilitated the progress of various downstream applications, including autonomous navigation and improved human-robot interaction.

Traditional SLAM system, such as the ORB-SLAM [1]–[3] and VINS [4] series, rely on feature extraction to achieve real-time tracking. However, the resulting sparse point maps are deficient in dense geometric and texture details, which limits their utility in supporting downstream tasks.

Recent advancements in high-fidelity reconstruction methods, such as Neural Radiance Fields (NERF) [5] and 3D Gaussian Splatting (3DGS) [6], have driven the exploration of end-to-end SLAM system. Pioneering works in this domain, including iMap [7] and NICE-SLAM [8], which use multi-layer perceptrons (MLPs) to learn neural implicit

representations of scenes. These approaches enable realistic reconstructions and predictions of unseen areas. However, NERF-based methods face significant limitations: they rely on ray casting for pixel rendering, which is computationally inefficient, and the overly smooth features learned by MLPs result in reconstructions lacking fine-grained details, thus constraining their application in real world.

In contrast, 3D Gaussian Splatting offers distinct advantages as explicit representation method, using rasterization for precise and fast rendering. Its explicit representation nature provides inherent scalability, positioning 3DGS SLAM as a promising alternative. Nonetheless, current research on 3DGS SLAM predominantly emphasizes improvements in rendering speed and reconstruction accuracy, often neglecting its potential as a comprehensive scene representation method capable of incorporating more information. To address this gap, researchers have explored incorporated semantic information into 3DGS SLAM, leading to the development of methods such as SGS-SLAM [9] and NEDS-SLAM [10]. These methods utilize masks and labels derived from semantic pre-segmentation to support the SLAM system. However, they fail to fully leverage the robust optimization potential of 3DGS.

To overcome this challenge, We enhances 3DGS by incorporating perceptual feature information extracted from vision models during training, aiming to achieve more accurate and essential semantic representations that assist SLAM.

Our main contributions are summarized as follows:

- We present SAP-SLAM, a 3DGS SLAM system that achieves precise semantic mapping by embedding perceptual features from vision models into the 3D Gaussian model. This method involves perceptual loss during training and optimization, which guides 3DGS to reconstruct scenes with improved accuracy and richer detail.
- We propose a novel semantic guidance Gaussian densification and pruning strategy that prioritizes complex local scenes based on the rendering difficulty of semantic information, thereby enhancing the overall quality of scene rendering.
- We conduct extensive evaluations on two challenging datasets, TUM-RGBD [11] and Replica [12], showing that our method not only delivers competitive performance in both scene reconstruction and tracking but also sets a new benchmark for efficiency in 3DGS SLAM systems.

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II. RELATED WORKS

A. Semantic SLAM

Semantic SLAM offers significant advantages by incorporating environmental information into SLAM systems through high-dimensional perceptual features, thereby enhancing a robot's understanding of unknown environments and facilitating downstream tasks [13]. However, effectively integrating semantic information with the underlying environmental representations to improve tracking and mapping remains a persistent challenge. Traditional SLAM methods with sparse 3D representations like point clouds, mesh [14]–[16] and Truncated Signed Distance Fields (TSDF) [17], [18] struggle to unlock the full potential of high-dimensional semantic understanding, hindering advancements in both reconstruction speed and model fidelity.

B. Neural Implicit SLAM

Neural Implicit representation has garnered significant attention for its high-fidelity scene reconstruction and differentiable rendering capabilities. Especially for SLAM, continuously differentiable scene representations provide substantial support for tracking and mapping optimization, enabling the development of end-to-end SLAM system. iMAP [7] pioneered the use of a single MLP to model entire scenes, achieving commendable tracking and reconstruction results. However, the limited capacity of a single MLP restricts its ability to capture fine details and leads to catastrophic forgetting in larger environments. NICE-SLAM [8] addresses these limitations by utilizing multiple pre-trained MLPs combined with geometric priors to create hierarchical feature representation. CO-SLAM [19] employs one-blob encoding, which merges the benefits of sparse encoding with enhanced local detail representation, to achieve more accurate performance in predicting unknown areas. GO-SLAM [20] integrates efficient loop closure detection and online global bundle adjustment, resulting in more consistent pose estimation. Collectively, these work have significantly propelled the progress of SLAM systems based on neural radiance fields.

C. 3D Gaussian Splatting SLAM

3D Gaussian representation have emerged as a promising approach for 3D scene reconstruction, thanks to its rapid rasterization and differentiable rendering capabilities. In contrast to NeRF-based SLAM, 3DGS employs direct parameter gradient flow to enable rapid optimization. This advancement not only enhances performance but also paves the way for more efficient SLAM solutions. SplatTAM [21] represents the scene using a set of simplified 3D Gaussians, utilizing visible contours to guide optimization and improve Gaussian densification in critical areas, thereby enhancing precision where it is most needed. GS-SLAM [22] introduces an adaptive expansion strategy that dynamically manages the addition or removal of 3D Gaussian components to efficiently reconstruct observed scene geometry. Gaussian-SLAM [23] segments the scene into independently optimized sub-maps, allowing it to scale effectively to larger scenes, thus addressing a major limitation of existing SLAM approaches.

Collectively, these methods demonstrate that 3DGS can achieve high tracking accuracy and high-fidelity reconstruction. However, the full potential of 3DGS as the foundational scene representation for SLAM is yet to be realized. By encoding semantic information directly into the 3DGS model, joint optimization can be utilized to create dense semantic fields, facilitating high-dimensional semantic understanding of complex scenes.

III. METHODS

An overview of the proposed method is presented in Figure 1. Sec.3.A outlines our semantic scene representation, detailing the extraction of semantic features and their embedding into the 3D Gaussian model following preprocessing. Sec.3.B describes our proposed semantic-guided densification and pruning strategies, which optimize poorly reconstructed areas, thereby enhancing system accuracy in adverse conditions. Sec.3.C and Sec.3.D elaborate on the SLAM process, encompassing tracking, mapping, and the associated loss functions.

A. Semantic 3D Gaussian Representation

To balance rendering speed and accuracy, we state a strategy similar to SplatTAM [21] for 3DGS representation as Equation 1, omitting the spherical harmonics component and using RGB colors $c_i = [r^i, g^i, b^i]^T$ as the DC component. Additionally, we assume that 3DGS is isotropic, which simplifies the model without significantly degrading fidelity.

$$g(\mathbf{x}) = o \exp\left(-\frac{\|\mathbf{x} - \mu\|^2}{2r^2}\right) \quad (1)$$

Here, $\mu \in \mathbb{R}^3$ represents the center of the Gaussian model, r is the Gaussian radius, and o indicates opacity.

The rapid differentiable rendering capability of 3DGS is a key factor in its effectiveness as a foundational representation in SLAM systems. For each independent 3D Gaussian model, rasterization maps the model onto the projection plane and pixelates it to generate synthetic views from various perspectives. The impact of each Gaussian on synthetic view can be rendered through point rendering formula as Equation 2.

$$\mu_{2D} = K \frac{T_i \mu}{d}, \quad r_{2D} = \frac{lr}{d}, \quad d = (T_i \mu) \quad (2)$$

where K is calibrated camera intrinsic matrix, T_i is camera pose at frame i , l is the focal length, d is the depth of Gaussian in camera coordinates. Following the sorting of Gaussian projections, color $C(p)$ and depth $D(p)$ are calculated using alpha blending, described as Equation 3 and Equation 4.

$$C(p) = \sum_{i \in N} c_i g_i(p) \prod_{j=1}^{i-1} (1 - g_j(p)) \quad (3)$$

$$D(p) = \sum_{i \in N} d_i g_i(p) \prod_{j=1}^{i-1} (1 - g_j(p)) \quad (4)$$

As a robust explicit representation method, 3D Gaussians can incorporate semantic information by adding additional

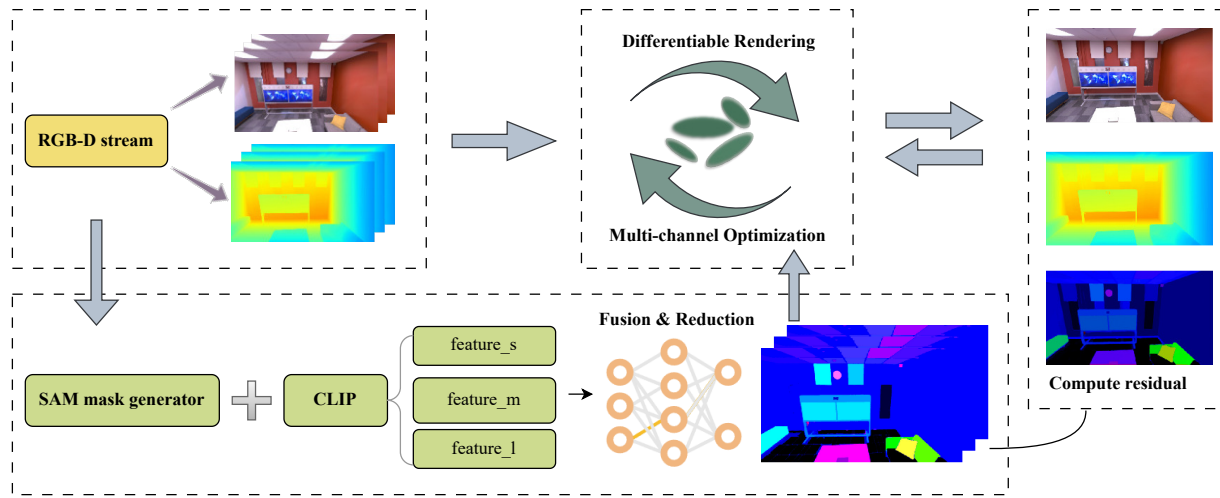


Fig. 1: **an overview of our SAP-SLAM.** For the input RGB-D stream, we segment it using the SAM model to obtain hierarchical masks, and then use the CLIP model to extract semantics features for each layer. The extracted semantics undergo dimensionality reduction via an MLP and are then subjected to a simplistic fusion process. This fused representation, in conjunction with photometric and depth maps, is used to optimize a Gaussian model. Furthermore, we calculate the residual between the rendered semantics and predicted semantics, which serves as guidance for the optimization process.

parameters to the 3DGS model. Obtaining three-dimensional semantic information directly for the entire scene is challenging; therefore, semantic SLAM typically extracts semantic features from 2D images using visual models and subsequently propagates them into the 3D space. Inspired by LangSplat [24], we employ the SAM [25] model for multi-layer semantic segmentation of images to generate three different levels of masks $\{M^s, M^m, M^l\}$, representing the subparts, parts, and overall of the scene, and use the CLIP [26] model to compute embeddings $\{f^s, f^m, f^l\}$ for each segmentation mask layer. Embedding the 512-dimensional feature vectors extracted by CLIP [26] directly into the 3DGS model would substantially increase the number of parameters, negating the advantage of 3DGS's rendering speed. To mitigate this, we train an autoencoder on the scene and use only the encoder to reduce the dimension of the CLIP [26] embeddings. The encoder-decoder consists of multiple linear layers and ReLU activation function. Its purpose is to reduce the dimensionality of semantic information to the same dimension as that of the 3D Gaussian DC component, thereby facilitating the semantic rendering process. The reduced-dimensional features are then embedded into the 3DGS model, and the rasterization process is modified to render semantic features $F(p)$ in a manner analogous to photometric rendering.

$$F(p) = \sum_{i \in N} f_i g_i(p) \prod_{j=1}^{i-1} (1 - g_j(p)) \quad (5)$$

It is important to emphasize that, unlike LangSplat [24], we do not render different semantic levels separately to create Gaussian models with hierarchical semantic information. Given that the objective of SLAM is to construct a

globally consistent map, we enhance the multi-level features through fusion. Our findings indicate that the fused features improve semantic classification and partially mitigate multi-view semantic inconsistencies. Additionally, this straightforward and effective fusion does not impose any additional computational burden on the optimization of 3DGS.

$$f(p) = f^s + f^m + f^l \quad (6)$$

B. Semantic-guided Gaussian Densification and Pruning Strategy

Neural radiance field-based SLAM systems improve camera pose estimation and reconstruction quality by comparing newly rendered views with input RGB-D frames and calculating photometric and depth losses. However, in 3DGS-based SLAM systems, reconstruction quality is even more critical due to gradient parameterization, which makes it a focal point of the optimization process. Reconstruction quality is heavily influenced by the number and distribution of 3D Gaussians; regions with poor reconstruction often result from insufficient Gaussian density or inadequate optimization. Additionally, artifacts and voids caused by outlier Gaussian points can significantly degrade tracking accuracy. Our experiments have shown that these poorly reconstructed regions correspond to higher noise responses in semantic rendering compared to well-rendered areas.

To tackle the challenges of poor reconstruction quality, we have developed a strategy that focuses on areas with suboptimal semantic rendering. This approach involves applying additional Gaussian densification and pruning in these regions, guiding the system to further optimize areas with low reconstruction quality. Specifically, we calculate the residual using L1 loss between reference semantics extracted

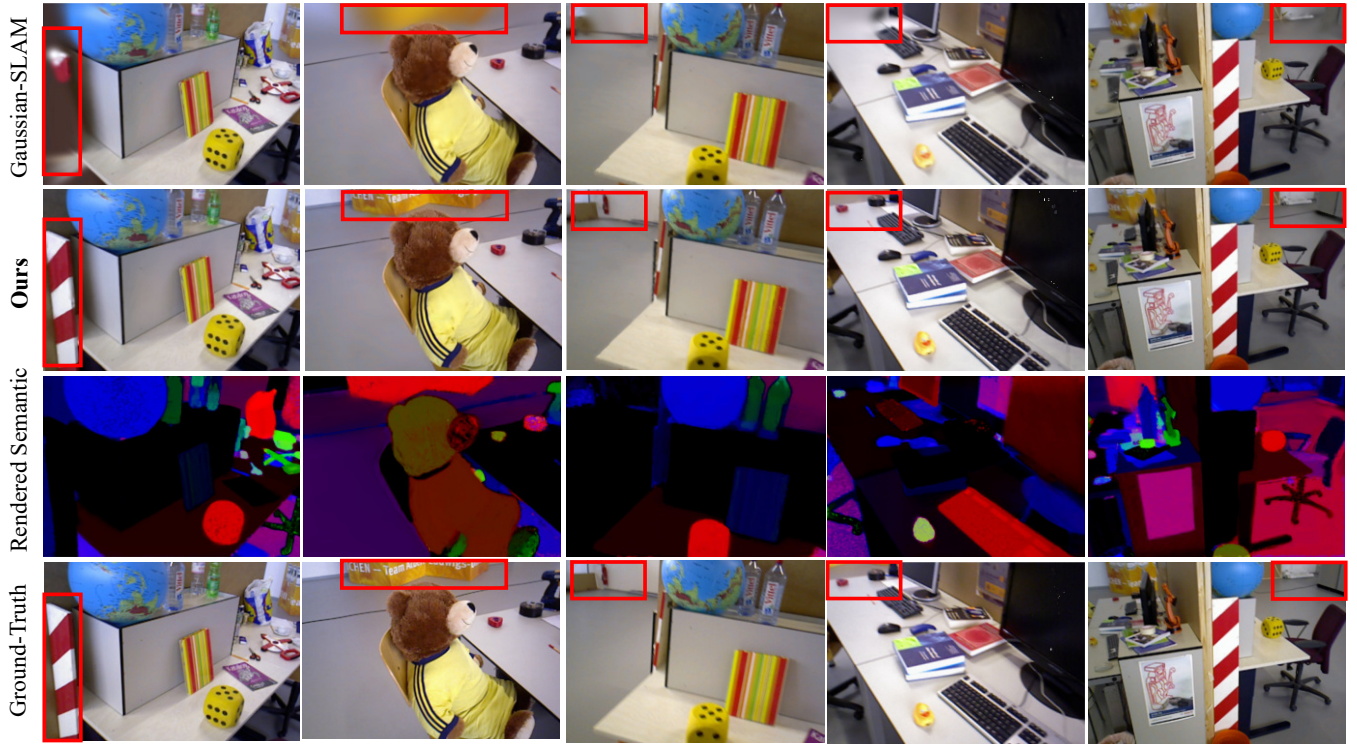


Fig. 2: Visualization between our method and the baseline Gaussian-SLAM. We visualize some scenes of TUM-RGBD, highlighting details with red boxes. When Gaussian SLAM faces fast movement and depth missing, the reconstruction effect is poor, while our method can still reconstruct normally.

from CLIP and the rendered semantics, summing the residual along the feature dimension to create a semantic noise distribution that matches the size of the rendered image. Pixel locations where the noise exceeds a threshold set above the median value are identified as targets for further Gaussian densification and optimization. Here we use the median depth error (MDE) as the threshold.

Recognizing that relying solely on semantic information to determine optimization regions may not be fully reliable, we extend our optimization criteria by incorporating an alpha threshold and a depth residual threshold as supplementary references to refine the optimization mask. We select regions with opacity below a certain threshold and depth residual greater than MDE to generate masks, and then merge them with semantic masks. This multi-model approach ensures that the optimization process is robust and comprehensive, effectively addressing areas of poor reconstruction. The final optimization mask, incorporating these thresholds, is defined as shown in Equation 7.

$$\begin{aligned}
 M(P) = & (\alpha(p) < T_\alpha) \\
 & + ((D_{GT}(p) < D(p)) (L_1(D(p)) > \lambda_1 MDE)) \\
 & + (M_f(L_1(F(p)) > \lambda_2 MDE))
 \end{aligned} \quad (7)$$

C. Tracking Process

During tracking, the camera pose T_{i+1} of the next frame is estimated using a constant velocity assumption based on the mapped scene.

$$T_{i+1} = T_i + (T_i - T_{i-1}) \quad (8)$$

Following this, we refine the estimated pose by iteratively minimizing the loss between the color and depth of the new view and the corresponding ground truth (GT) values. Consistent with the approach used in Gaussian-SLAM, all Gaussian parameters are kept frozen during the optimization process. Additionally, alpha and depth error masks are applied to mitigate the impact of poorly reconstructed regions on pose estimation. We use L1 loss for both L_c and L_d in tracking. Similar to Equation 7, we adopt opacity weighting and discard regions with depth errors greater than the threshold to generate a guidance mask M_t . The loss function is defined as follows:

$$L_{tracking} = M_t (\lambda_{t_c} L_c + \lambda_{t_d} L_d) \quad (9)$$

It is important to note that semantic-related constraints were not included in the tracking loss because semantic predictions tend to exhibit greater variability compared to color and depth rendering. Such inconsistencies can negatively influence the tracking loss, which is particularly sensitive to deviations. Therefore, it is more effective to compute the tracking loss using regions that exhibit higher certainty and superior reconstruction quality.

D. Mapping Process

During mapping, Gaussians are added for new scenes entering the view frustum, and the Gaussian parameters of

TABLE I: Quantitative comparison of our method and the baselines in training view rendering on the Replica dataset. Our results have achieved competitive performance. **SOTA** performances are highlighted.

Methods	Metrics	Room0	Room1	Room2	Office0	Office1	Office2	Office3	Office4	Avg.
NICE-SLAM	PSNR↑	22.12	22.47	24.52	29.07	30.34	19.66	22.23	24.94	24.42
	SSIM↑	0.689	0.757	0.814	0.874	0.886	0.797	0.801	0.856	0.809
	LPIPS↓	0.330	0.271	0.208	0.229	0.181	0.235	0.209	0.198	0.233
Point-SLAM	PSNR↑	32.40	34.08	35.50	38.26	39.16	33.99	33.48	33.49	35.17
	SSIM↑	0.974	0.977	0.982	0.983	0.986	0.960	0.979	0.975	0.975
	LPIPS↓	0.113	0.116	0.111	0.100	0.118	0.156	0.132	0.142	0.124
SplaTAM	PSNR↑	32.86	33.89	35.25	38.26	39.17	31.97	29.70	31.81	34.11
	SSIM↑	0.98	0.97	0.98	0.98	0.98	0.97	0.95	0.95	0.97
	LPIPS↓	0.07	0.10	0.08	0.09	0.09	0.10	0.12	0.15	0.10
Gaussian-SLAM	PSNR↑	38.88	41.80	42.44	46.40	45.29	40.10	39.06	42.65	42.08
	SSIM↑	0.993	0.996	0.996	0.998	0.997	0.997	0.997	0.997	0.996
	LPIPS↓	0.017	0.018	0.019	0.015	0.016	0.020	0.020	0.020	0.018
SGS-SLAM	PSNR↑	32.50	34.25	35.10	38.54	39.20	32.90	32.05	32.75	34.66
	SSIM↑	0.976	0.978	0.981	0.984	0.980	0.967	0.966	0.949	0.973
	LPIPS↓	0.070	0.094	0.070	0.086	0.087	0.101	0.115	0.148	0.096
NEDS-SLAM	PSNR↑	35.23	34.86	35.16	37.53	39.71	32.68	31.07	31.82	34.76
	SSIM↑	0.979	0.862	0.983	0.981	0.979	0.973	0.968	0.973	0.962
	LPIPS↓	0.082	0.075	0.071	0.091	0.087	0.079	0.103	0.113	0.088
Ours	PSNR↑	39.20	41.74	42.64	46.24	45.60	39.79	39.12	42.81	42.14
	SSIM↑	0.993	0.995	0.997	0.997	0.997	0.996	0.996	0.997	0.996
	LPIPS↓	0.019	0.018	0.019	0.012	0.014	0.020	0.020	0.017	0.017

TABLE II: Quantitative comparison of our method and the baselines in tracking performance on the Replica dataset. We use Absolute Trajectory Error(ATE) RMSE(cm)↓ for metrics. **SOTA** performances are highlighted.

methods	Room0	Room1	Room2	Office0	Office1	Office2	Office3	Office4	Avg.
NICE-SLAM	1.69	2.04	1.55	0.99	0.90	1.39	3.97	3.08	1.95
Point-SLAM	0.61	0.41	0.37	0.38	0.48	0.54	0.72	0.63	0.52
SplaTAM	0.31	0.40	0.29	0.47	0.27	0.29	0.32	0.55	0.36
Gaussian-SLAM	0.29	0.29	0.22	0.37	0.23	0.41	0.30	0.35	0.31
SGS-SLAM	0.46	0.45	0.29	0.46	0.23	0.45	0.42	0.55	0.41
NEDS-SLAM	0.37	0.40	0.33	0.35	0.28	0.30	0.32	0.47	0.35
Ours	0.30	0.26	0.22	0.51	0.22	0.38	0.23	0.22	0.29

existing scenes are optimized. Following the approach in Gaussian-SLAM [23], the input RGB-D sequence is divided into multiple sub-scenes. A new sub-scene is created when the translation or rotation of the current frame relative to the first frame of the existing sub-scene exceeds a predefined threshold. For the first keyframe of a sub-scene, a dense point cloud is generated through the back-projection of RGB values, and Gaussians are added by uniformly sampling in regions with high color gradients. For subsequent keyframes, we use the mask obtained from the method described in Equation 7 to identify locations where new Gaussians should be added. In these regions, Gaussians are uniformly sampled while ensuring no other Gaussians exist within a radius of ρ .

During the optimization process, we jointly optimize color, depth, and semantic information with the goal of minimizing the following loss function:

$$L_{mapping} = \lambda_{m_c} L_c + (1 - \lambda_{m_c}) L_{ssim} + \lambda_{m_d} L_d + \lambda_{m_{reg}} L_{reg} + \lambda_{m_f} L_f \quad (10)$$

For the color loss, a weighted combination of L1 and SSIM losses is used, while the depth loss is directly computed using the L1 loss. Additionally, isotropic regularization is applied to penalize Gaussians that become excessively elongated.

For the semantic loss, we use the L2 loss, which demon-

strates better convergence within a limited number of iterations. The semantic mask calculated in Equation 7 effectively reduces the impact of outliers.

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We evaluate the performance of SAP-SLAM on both synthetic and real-world datasets with semantic maps. Following other NeRF-based and 3DGS SLAM methods, we assessed our tracking and reconstruction performance on 8 synthetic scenes from Replica and 3 real-world scenes from TUM-RGBD.

Metrics. We employ several metrics to evaluate the reconstruction quality. For tracking accuracy evaluation, we use Absolute Trajectory Error(ATE). The quality of the 3D reconstructions is assessed using PSNR, SSIM, and LPIPS.

Baselines. Our approach was compared against several state-of-the-art (SOTA) dense visual SLAM, including NeRF-based SLAM methods such as NICE-SLAM, CO-SLAM, and Point-SLAM, in addition to 3DGS SLAM approaches like Splatam and Gaussian-SLAM. For comparison of semantic methods, we consider SGS-SLAM, NEDS-SLAM as baseline.

Implementation Details. The SLAM system was implemented on a single NVIDIA RTX 3090 GPU. For the Replica dataset, mapping is performed every 5 frames, with

TABLE III: Quantitative comparison of our method and the baselines in training view rendering on the TUM-RGBD dataset. Our method performs well in real-world scenes, especially in situations where there is a significant gap in depth compared to baseline methods. SOTA performances are highlighted.

Methods	Metrics	fr1/desk	fr2/xyz	fr3/office	Avg.
NICE-SLAM	PSNR↑	13.83	17.87	12.590	14.86
	SSIM↑	0.569	0.718	0.554	0.614
	LPIPS↓	0.482	0.344	0.498	0.441
Point-SLAM	PSNR↑	13.87	17.56	18.43	16.62
	SSIM↑	0.627	0.708	0.754	0.696
	LPIPS↓	0.544	0.585	0.448	0.526
SplaTAM	PSNR↑	22.00	24.50	21.90	22.80
	SSIM↑	0.857	0.947	0.876	0.893
	LPIPS↓	0.232	0.100	0.202	0.178
Gaussian-SLAM	PSNR↑	24.01	25.02	26.13	25.05
	SSIM↑	0.924	0.924	0.939	0.929
	LPIPS↓	0.178	0.186	0.141	0.168
Ours	PSNR↑	25.19	23.61	26.83	25.21
	SSIM↑	0.939	0.926	0.946	0.937
	LPIPS↓	0.141	0.186	0.108	0.145

100 iterations of optimization conducted during the mapping phase and 60 iterations during the tracking phase. For the TUM-RGBD dataset, mapping is conducted for every frame, with 100 optimization iterations during the mapping phase and 200 iterations during the tracking phase. The weighting coefficients of each loss are $\lambda_{t_c} = 0.6$, $\lambda_{t_d} = 0.4$ in tracking, $\lambda_{m_c} = 0.8$, $\lambda_{m_d} = \lambda_{m_{reg}} = 0.8$, $\lambda_{m_f} = 0.02$ in mapping.

B. Experiment results

We present the rendering results on the Replica and TUM-RGBD datasets in Table I and Table III, along with quantitative comparisons to other baseline methods. Our method achieves a higher rendering performance compared to existing approaches. Existing semantic SLAM systems often do not conduct experiments on the TUM-RGBD dataset. This may be due to the lack of ground truth semantic labels and challenges associated with missing depth data in some scenes. Despite these challenges, our method performs well and even achieves SOTA performance, suggesting that our proposed semantic approach effectively targets poorly reconstructed areas and enhances overall reconstruction quality.

Figure 2 provides a visual comparison between our method and the baseline Gaussian-SLAM on the TUM-RGBD dataset. In regions that have just entered the view frustum or in areas with missing depth data, Gaussian-SLAM experiences latency in updating depth information, resulting in blurred reconstructions. In contrast, our method, which leverages semantic information, reduces reconstruction errors by partially recovering these failed reconstructions, thereby mitigating their negative impact on tracking accuracy.

Table II and Table IV present the tracking and reconstruction results of our method on the Replica and TUM-RGBD datasets. Compared to baseline methods, our approach demonstrates a slight improvement in tracking accuracy. Nevertheless, our method exhibits suboptimal performance in specific scenarios, likely due to the introduction of excessive Gaussians in poorly reconstructed regions. This

TABLE IV: Quantitative comparison of our method and the baselines in tracking performance on the TUM-RGBD dataset. We use ATE RMSE(cm)↓ for metrics. Adding too much Gaussian in real scenes may have an impact on tracking, but it still demonstrates competitive performance. SOTA performances are highlighted.

Methods	fr1/desk	fr2/xyz	fr3/office	Avg.
NICE-SLAM	4.3	31.7	3.9	13.3
Point-SLAM	4.3	1.3	3.5	3.0
SplaTAM	3.4	1.2	5.2	3.3
Gaussian-SLAM	2.6	1.3	4.6	2.9
Ours	2.6	1.3	3.9	2.6

TABLE V: Ablation study on desk and room0 scenes. F_loss means Feature Loss and S_mask means Semantic-guided Mask.

F_loss	S_mask	fr1/desk		room0	
		ATE RMSE↓	PSNR↑	ATE RMSE↓	PSNR↑
✗	✗	3.0	24.01	0.44	38.47
✓	✗	2.8	24.93	0.31	38.91
✗	✓	2.9	24.41	0.32	38.55
✓	✓	2.6	25.19	0.30	39.20

over-addition, without adequate optimization, may introduce noise that negatively impacts trajectory estimation.

C. Ablation Study

Table V shows the results of ablation experiments conducted on desk scene from the TUM-RGBD dataset to assess the effectiveness of the semantic embedding loss and the semantic-guided densification strategy. Incorporating the semantic embedding loss resulted in an obvious improvement in rendering and tracking performance. This improvement is due to semantic optimization, which allows the model to understand the relationships between objects in the scene without altering the Gaussian distribution. Furthermore, the introduction of the semantic-guided densification strategy significantly increased the number of Gaussian points in the scene, enhancing opportunities for densification and optimization in regions that previously had poor reconstruction quality. As a result, the overall reconstruction quality has been improved.

V. CONCLUSIONS

We present an end-to-end semantic SLAM system based on 3D Gaussian splatting, integrating hierarchical semantic features from vision pre-trained models into the 3DGS framework. Our approach enables high-fidelity rendering, robust camera tracking, and accurate 3D semantic reconstruction. The proposed semantic fusion and semantic-guided Gaussian densification strategies focus the models attention on areas with low reconstruction quality, facilitating targeted optimization to enhance overall SLAM performance.

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